



UNIVERSITAT
Carlemany

Formación europea
para desafíos reales

De:

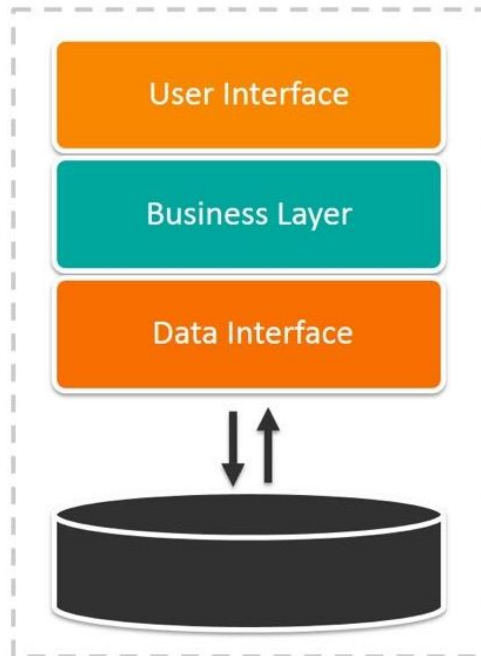
 Planeta Formación y Universidades

CLOUD

Alternative databases

1. Motivation
2. Object storage
3. In memory

Monolithic Architecture

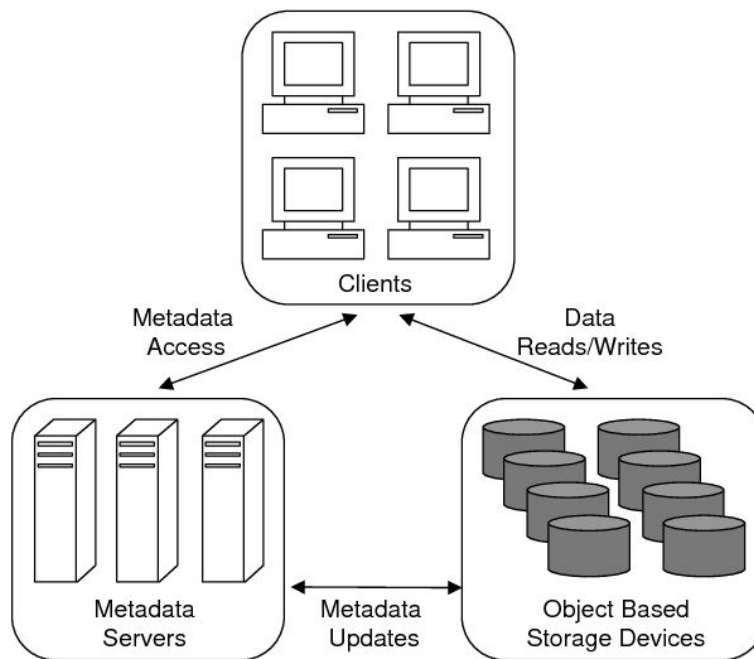


- **A**tomicity
- **C**onsistency
- **I**solation
- **D**urability

Definition

Object storage, also known as object-based storage, is a computer data storage architecture designed to handle large amounts of unstructured data. Unlike other architectures, it designates data as distinct units, bundled with metadata and a unique identifier that can be used to locate and access each data unit.

Architecture



- S3: managed object storage service AWS
- Store static files
 - ML models
 - Files
 - Websites
 - Data lakes
 - Backups
- 99.999999999% durability (11 9's)



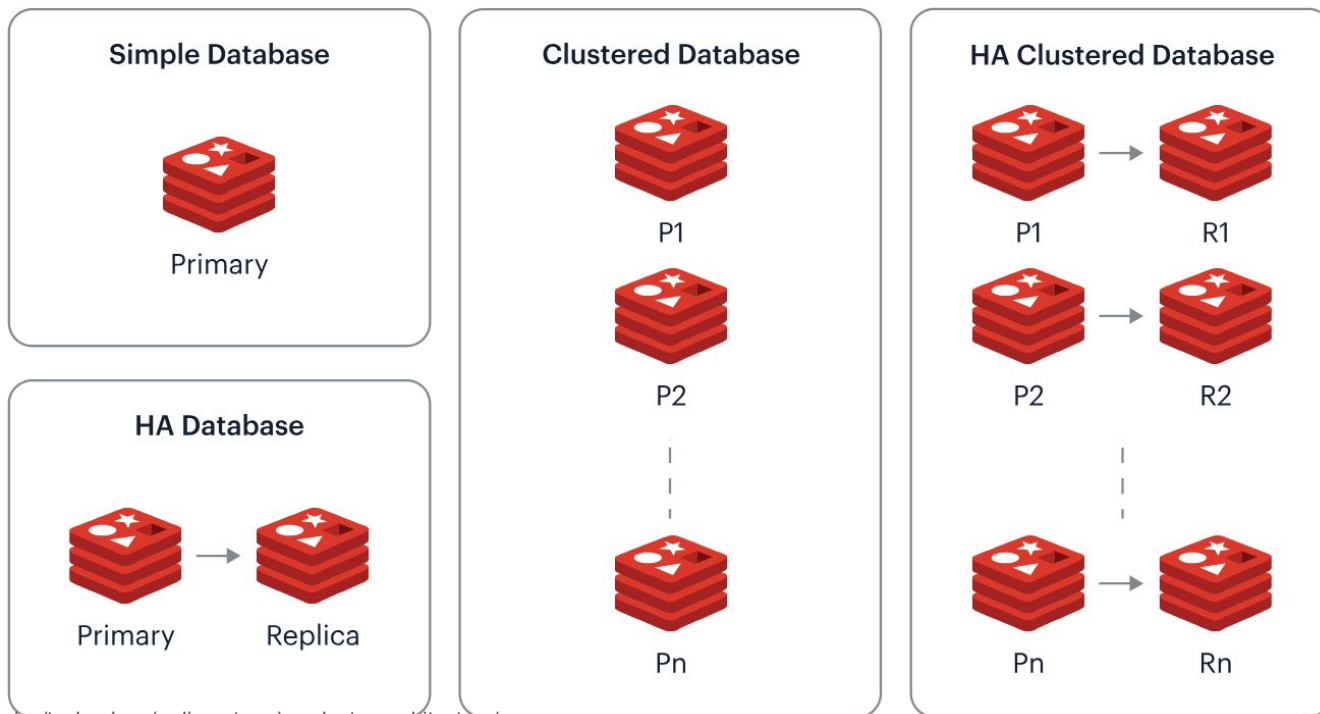
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Definition

An in-memory cache relies on random access memory (RAM) to store data. The technology stores data tables directly in the RAM instead of on external drives. Specialized data structures allow the indexing of your data records. The indexes act as direct pointers to specific rows and columns. However, the actual physical data is compressed and non-relational in format. When you make an access request, the database navigates to the exact data value using the index. Stored data is always available in a directly usable format.

Architecture





- ElastiCache: managed in memory key-value storage service AWS
- Can store backups in S3
- Queue service
- Real time analysis